FICTIONAL ELECTRICAL SERVICES

Home energy connection schedule

PV inverter, AC distribution and EV wallbox

Site reference	HOME-001
Premises	12 Example Avenue, Budapest 1111
Commissioning date	18 January 2026
Revision	As-built issue 1

As-built description of the fictional home's grid supply, solar inverter, metering, protective devices and installed EV charging circuit.

Read the identification, safety and applicability information before relying on this document.

2. System topology

Energy path

PV strings connect to the HelioSync I10 inverter. The inverter converts PV DC to three-phase AC and connects to the main distribution board.

Wallbox path

The GridNest W11 wallbox is supplied from a dedicated AC circuit at the main distribution board. The wallbox is not connected directly to PV DC conductors.

Grid connection

The public grid and inverter share the AC bus. When enabled, grid import supplements PV output for household and EV loads.

From	Through	To
PV array	HelioSync I10 inverter	Main AC distribution board
Main AC board	Dedicated protection and cable	GridNest W11 wallbox
Grid connection	Main meter and board	Household AC bus

There is no permitted direct panel-to-vehicle path in this installation.

3. Circuit schedule

Available wallbox rating

EV-01 is commissioned for 16 A per phase, corresponding to approximately 11 kW three-phase charging.

Load management

The smart meter reports all three phases so the energy manager can reduce EV current before the 3 x 25 A service limit is exceeded.

Circuit	Supply	Protection	Cable	Connected equipment
EV-01	3-phase 400 V	3 x 16 A breaker; Type A RCD; wallbox 6 mA DC detection	5 x 4 mm2 copper	GridNest W11, GN-W11-01
PV-AC	3-phase 400 V	3 x 20 A breaker and AC isolator	5 x 4 mm2 copper	HelioSync I10, HS-I10-3P
MAIN	3-phase 400 V	3 x 25 A service limit	Utility service	Main distribution board

4. Metering and communications

Smart meter

A three-phase meter at the grid connection reports import, export and phase current to the HelioSync gateway.

Network

The inverter gateway and GridNest W11 are connected by wired Ethernet on the local energy-control network.

Control profile

The commissioned wallbox control profile is HS-EV-2 with a five-second nominal update interval.

Fail-safe

On communication loss the W11 current is limited to 6 A per phase until valid measurements return.

Device	Address/role
HelioSync I10	Energy manager and inverter gateway
GridNest W11	Flexible EV load
Three-phase meter	Grid import/export and phase current

5. Commissioning record

Identity check

Installed inverter HS-I10-3P and wallbox GN-W11-01 match their supplied manuals. The Orion E4 vehicle was used for the functional charging test.

Electrical test

Phase-to-phase voltage measured 399-402 V. Protective-earth continuity, RCD operation and phase sequence passed.

Charging test

The wallbox delivered 15.8-16.0 A per phase, approximately 10.9 kW, before load management. The vehicle accepted the session without a rating conflict.

Energy management test

With simulated household load, the controller reduced wallbox current and restored it when capacity returned. Grid import was enabled.

Installed device	Model/code	Result
EV	Orion E4 / ORI-E4-77	11 kW AC accepted
Wallbox	GridNest W11 / GN-W11-01	Commissioned at 16 A/phase
Inverter	HelioSync I10 / HS-I10-3P	Gateway communication passed

6. Limitations and handover

Installed equipment only

Ratings from an alternative wallbox, inverter or vehicle brochure do not describe the installed system unless the connection schedule is revised after replacement.

No direct PV charging

The system requires inverter conversion, the AC distribution and protective devices, and the wallbox control function.

Charge-time estimates

Actual time depends on vehicle SoC, target, temperature, household load and energy-management policy.

Document set

Keep this schedule with the Orion E4 guide, GridNest W11 guide and HelioSync I10 manual.

Site relation	HOME-001
Source ID	s6-electrical-connection-description
Document status	Synthetic as-built evaluation record
Revision	Issue 1 - 18 January 2026